

Development of a Numerically Controlled Hot Wire Foam Cutting Machine for Wing Mould Construction

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Abstract—The foam mould making for Unmanned Aerial Vehicles should be efficient and accurate to improve research and manufacturing. Hot wire cutting is a widely used method in foam cutting. In hot wire cutting, accuracy and quality of the foam cut mainly depends on the variable cutting parameters which affect the cutting process. To perform a proper cut with required quality the cutting parameters should be set precisely and accurately. This research is done to identify the variations and inter dependency of cutting parameters. It is important to estimate appropriate cutting parameter values before the actual cut. The results of this research aid to improve the hot wire foam cutting by solving the limitations and drawbacks to select best cutting parameters of the Computer Numerically Controlled machine which is in progress.

Keywords - *cnc foam cutting machine, hot wire cutting, foam cutting parameters, Wing mould making machine*

I. INTRODUCTION

CNC hot wire foam cutting machines are used to cut solid shapes using foam, autonomously controlled by the commands obtained by computational model. The industry of Unmanned Aerial Vehicles (UAV) in Sri Lanka is lacking an effective method to produce components like wing and fuselage. The conventional method of making fibreglass wings is using a wooden plug. It is experienced the method does not yield sufficient surface finish in the outer surface. If the airplane parts are produced inside moulds, high accuracy and surface quality can be acquired [1]. Construction of foam moulds to manufacture these components can be identified as a good solution [2]. However, proper methods should be implemented to cut foam materials with the required accuracy. CNC hot wire foam cutting is an effective way of producing foam products with high accuracy and surface finish [3].

Development of a CNC hot wire foam cutting machine with required performance should be done paying attention to several factors. The attention should be given to the existing machines and their mechanisms [4] [5] [6] [7]. The literature emphasizes that there are limitations and drawbacks in hot wire foam cutting process [7]. New methods should be

explored to overcome those limitations by controlling the parameters involved in the cutting process.

Identifying and controlling the cutting parameters are considered as the main factors affecting the product quality. The controlling should be precisely done to move the hot wire in the correct path [8]. Errors which occur when selecting and setting the cutting parameters will result in a faulty product. That is not compatible with its intended application.

Precise control for the cutting parameters, proper mechanisms and controllers should be implemented by analysing the behaviour and characteristics of cutting parameters. There are quite a few research papers published in this area. The mechanisms used in advanced foam cutting machines are not published public. The main cutting parameters involved are cutting wire material and its dimensions, cutting speed, cutting temperature, tension of the cutting wire and kerf width.

Considering the existing mechanisms, the best method has to be selected and the improvements must be done to get the best possible output. In this study, a set of experiments were carried out to find these conditions and methods. After conducting the experiments on Styrofoam (Polystyrene (PS)) the results were analysed. By plotting the obtained data, the best conditions for conducting the cutting operation were identified. A feed rate equation to obtain the constant kerf width with respect to variable temperature was developed by plotting multidimensional relationships between different parameters.

II. METHOD

In order to find out the appropriate cutting parameters and conditions it is required to carry out experiments. This section explains the identified cutting parameters and experiment setup that used in this study.

Out of the two cutting mechanisms available pure thermal cutting was selected for conducting the experiment.

A. Cutting parameters on hot wire cutting

Foam work pieces were cut in different cutting conditions to find out appropriate cutting conditions and the variation of properties of the cut with compelling parameters. The cutting parameters of hot wire foam cutting are hot wire material and its dimensions, cutting speed, cutting temperature and kerf width [7]. Kerf width is the width of the gap between foam parts which are separated by the cut [8]. The cutting conditions were changed by varying the pre-adjustable cutting parameters; cutting temperature, cutting speed and wire diameter.

Mainly there are two types of hot wire foam cutting methods; thermo mechanical shearing method and pure thermal cutting method [8]. In thermo mechanical shearing, hot wire touches the foam and applies force to shear out foam material. In pure thermal cutting, hot wire does not touch foam material and the foam vaporizes due to heat, forming the cut.

B. Experiment Setup

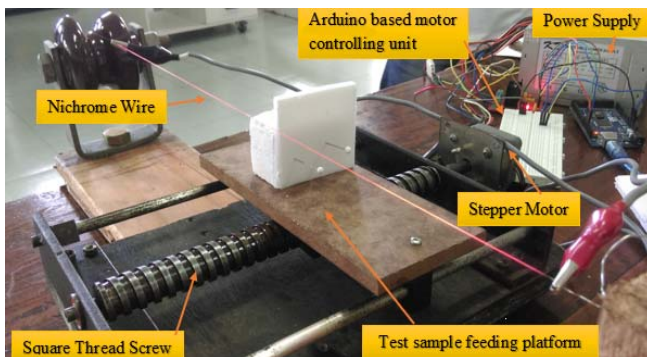


Fig.1 Experiment setup

The experiment set used in this study is depicted in Fig.1. It mainly consists of Nichrome wire tensioning mechanism, feed rate control mechanism and current controlling apparatus. Foam work pieces were mounted on a platform and fed into the hot wire in a constant velocity. Heating of wire was done by current which was supplied by connecting the two wire end to a power supply unit.

The wire tensioning mechanism is consisted of a Nichrome wire which was tensioned by weights hanging at one end of wire. The standards weights of 500g were used to tension the wire. The foam feeding unit consisted of a square threaded screw drive of 6mm pitch coupled with a stepper motor (NEMA 23) which was driven by a by an embedded systems (Arduino Mega 2560) based Pulse Width Modulation (PWM) signals aided with stepper motor controller and 12V power supply. Foam work pieces were mounted on a platform which was connected and driven by rotating screw. Current was supplied to the wire by a dual DC power supply (King Instrument Electronics co. Ltd). A digital thermocouple thermometer (Omega CL3515) was used to measure the temperature of wire. Since the cutting temperature was needed to be measured the thermocouple node was fixed to the wire 5mm away from the point of cutting.

The tension of the wire was determined by the number of standard weights used. Bench top power supply was used to measure and regulate the current supply to the hot wire. Feed rate was calculated using the step count of the motor and the screw drive's pitch.

Experiment was carried out using two Nichrome wires with the diameters 0.6mm and 0.32mm which were the only wire sizes available in Sri Lankan market. Typically, wire diameter range from 0.2mm to 0.5mm is convenient for hot wire foam cutting [7].

The first experiment was conducted to select a proper wire for the foam cutting process. The temperature variation of the wire with respect to the supply current was measured. The kerf width was also examined using foam samples. The next step was to obtain a matrix of test samples with respect to different feed rates and temperatures. The kerf width of the samples were accurately measured using a Gear Profile microscope. From this machine, length measurements can be taken in the accuracy of 0.001mm. Then the resulting data were analysed.

C. Specification of Test Samples

The experiments were conducted using Styrofoam (poly(1-phenylethylene)), which is used in mould making process. The density of the sample was 1.04 g/cm³. Melting point was 240 °C and thermal conductivity 0.033 W/ (m K).

D. Variation of temperature with current

When the current is applied to the wire, it heats up and comes to a stable temperature after a certain amount of time. That stable temperature was measured for different current amounts. This is done before the cutting process to find out the current value to acquire particular temperature value.

This experiment was carried out for Nichrome wires with diameters 0.6mm and 0.32mm. These are the wire types which are commonly available. The wire diameter range from 0.2mm to 0.5mm is convenient for hot wire foam cutting [7]. 0.6mm diameter wire can withstand higher tensions than 0.32mm wire and 0.6mm wire was used for this test because wire sagging can be easily avoided with that.

TABLE I. WIRE TEMPERATURE VS CURRENT FOR 0.32MM WIRE

Current(A)	Wire temperature(°C)
0.5	42
1.0	85
1.5	175
2.0	250
2.2	255
2.4	310
2.5	330
2.7	375
3.0	420

TABLE II. WIRE TEMPERATURE VS CURRENT FOR 0.6MM WIRE

Current(A)	Wire temperature(°C)
0.5	29
1.0	45
1.5	66
2.0	98
2.5	125
2.6	137
2.8	148
2.9	158

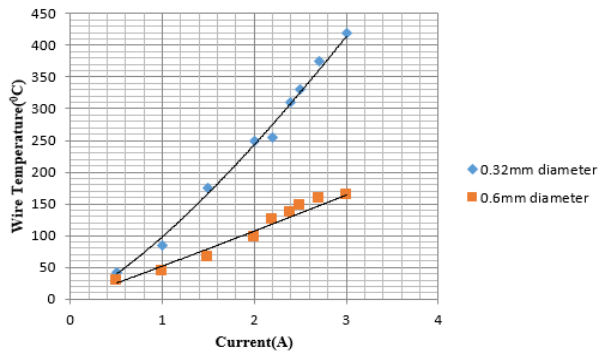


Fig.2 Wire temperature vs current

Fig.2 illustrates the variation of wire temperature of two wire types when the supplied current varies. Wire temperature increases with time. The temperature of 0.32mm diameter wire increases faster rate than 0.6mm diameter wire.

For pure thermal cutting, the wire should be heated over 300°C [9]. Fig.2 depicts that 0.32mm diameter wire can reach this temperature with less amount of current.

From here onwards, the experiments were carried out only for the pure thermal cutting. Therefore only 0.32mm diameter wire was used because it gives high temperatures with less power requirement. Moreover, this wire diameter is within the appropriate wire cutting diameter range 0.2mm to 0.5mm.

E. Variation of temperature with time (when no cutting is done)

0.32mm diameter wire was used for this test. Current was supplied to the wire and temperature was measured until it came to the stable temperature value. No cutting has been performed when taking the readings shown in Table III.

TABLE III. WIRE TEMPERATURE VS TIME

Time(s)	Wire temperature(°C)	Time(s)	Wire temperature(°C)
0	25	11	285
1	28	12	291
2	41	13	296
3	69	14	301
4	110	15	304
5	160	16	309
6	187	17	312
7	202	18	310
8	228	19	314
9	255	20	311
10	273		

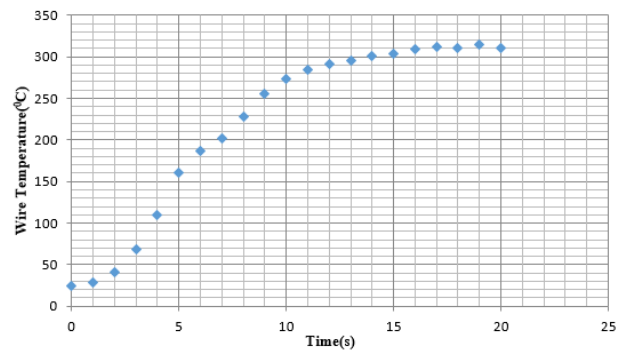


Fig.3 Wire temperature vs time when no cutting is performed

Fig.3 illustrates the variation of wire temperature with the time when cutting is not done. Initially the wire temperature increases gradually. Then it reaches to a stable value.

F. Variation of temperature with time during the cutting process

This section of the experiment was performed only for pure thermal cutting. To attain pure thermal cutting, the wire should be heated over 300°C. Therefore over 2.5A current should be applied to the wire to get to that temperature according to the result of section A of the experiment.

2.5A current was applied to the wire and the cutting was started after the wire has come to the stable temperature. Temperatures and time in Table IV were recorded while the cutting is done.

TABLE IV. WIRE TEMPERATURE VS TIME DURING CUTTING PROCESS

Time(s)	Wire temperature(°C)	Time(s)	Wire temperature(°C)
0	317	11	283
1	314	12	280
2	307	13	289
3	304	14	297
4	303	15	295
5	300	16	267
6	296	17	275
7	293	18	275
8	288	19	271
9	280	20	273
10	278	21	276

Fig.4 illustrates the variation of wire temperature with time when the cutting is performed. At the beginning the wire

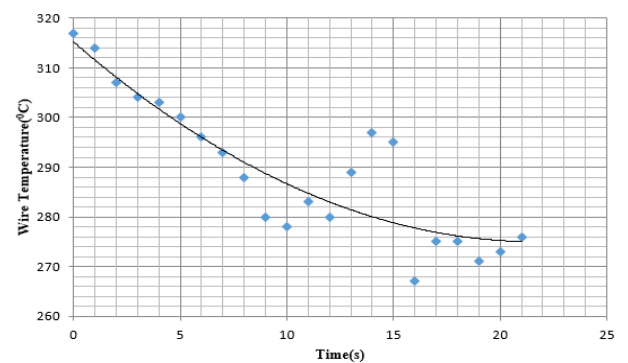


Fig.4 Wire temperature vs time during the cutting process

temperature is at maximum when there is no heat transferring. It consumes a little time for the temperature of the wire to reach steady state. To get uniform properties and the accuracy in the product, the product should be cut after stable cutting conditions and the stable temperature has arrived.

G. Variation of temperature with cutting distance

This section of the experiment was performed only for pure thermal cutting. Therefore 2.5A current was applied to the wire and the cutting was started after the wire has come to the stable temperature. Temperatures at different positions of the cutting path were recorded.

TABLE V. WIRE TEMPERATURE VS CUTTING DISTANCE

Cutting distance(cm)	Wire temperature(°C)
0	317
1	309
2	303
3	298
4	294
5	288
6	280
7	290
8	276
9	267
10	277

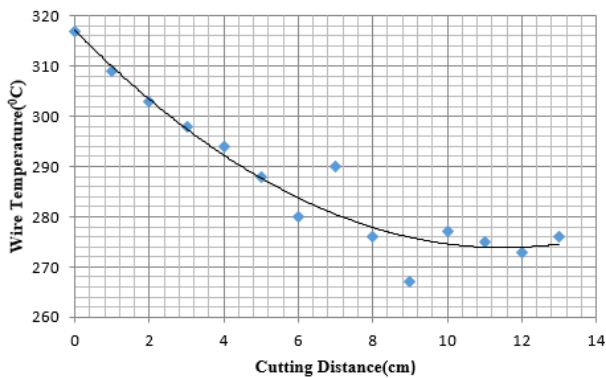


Fig.5 Wire temperature vs cutting distance

In Fig.5, the variation of wire temperature with cutting distance is shown. The highest value of cutting temperature is at the beginning of the cutting process. With cutting distance the temperature reduces and comes to a stable value. To get uniform properties and the accuracy in the product, the product should be cut after stable cutting conditions. Therefore some part of the foam material has to be cut before the cutting process of the required part.

H. Variation of kerf width with current

Kerf width is the width of the gap between foam parts which are separated by the cut. It is shown in Fig.6.

When the current through the hot wire varies, the temperature of the wire varies. When the temperature of the wire varies, the kerf width of the cut varies. Foam samples were cut using different current values. The feed rate was kept

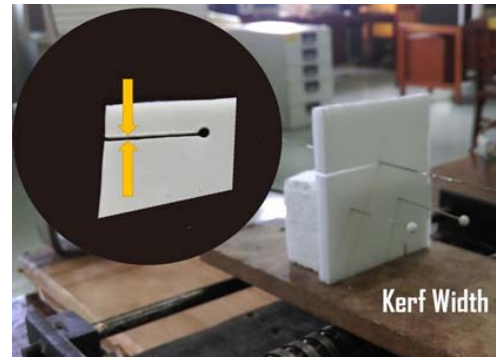


Fig.6 Kerf width

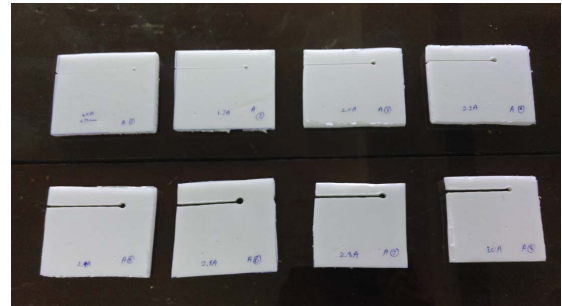


Fig.7 Test pieces

constant at the speed of 20.83mm/s. Then the kerf widths of those samples were measured. The foam samples are shown in Fig.7.

TABLE VI. KERF WIDTH VS CURRENT

Current (A)	Initial Temperature (°C)	Kerf width (mm)
1.5	175	0.330
1.7	210	0.336
2.0	250	0.410
2.2	260	0.674
2.4	310	1.192
2.6	330	1.530
2.8	375	1.582
3.0	410	1.840

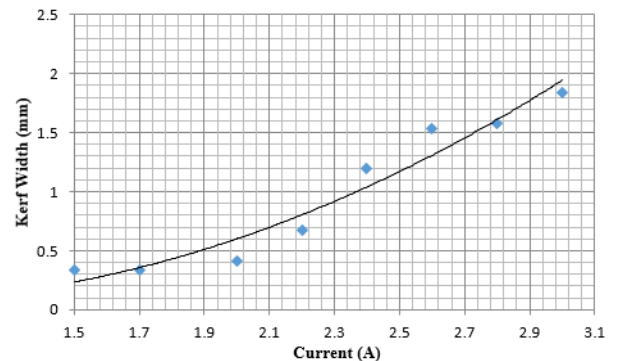


Fig.8 Kerf width vs current

According to the graph shown in Fig.8, kerf width increases with current. When current increases, the heat generated from the hot wire increases and more material from the foam vaporizes creating wider kerfs.

I. Variation of kerf width with feed rate

When feed rate varies, the amount of heat dissipated to the foam material varies. This varies the amount of material vaporization from the foam creating different kerf widths. The experiment was carried out by applying constant current of 2.0A to the cutting wire. Foam test pieces were cut using different feed rates and the kerf widths were measured.

TABLE VII. KERF WIDTH VS FEED RATE

Feed rate (mm/s)	Kerf width (mm)
6.25	2.57
8.33	1.66
10.42	1.51
12.5	1.24
14.58	1.16
16.67	0.91
18.75	0.9
20.83	0.8

Fig.9 illustrates the variation of kerf width with the feed rate.

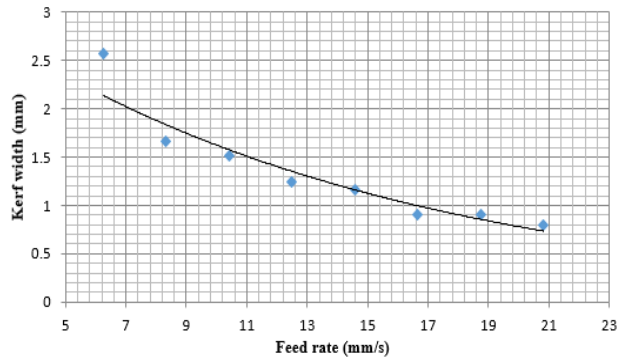


Fig.9 Kerf width vs feed rate

It can be concluded that in low feed rates, the kerf width is large. Kerf width reduces when feed rate increases. In low cutting speeds, there is much time to dissipate heat to vaporize foam.

The analysis proved that higher temperatures can be achieved efficiently using wires with smaller diameters. 0.6mm diameter wire requires large currents to achieve high temperature. The wires with low diameters are more suitable for pure thermal cutting.

The cutting temperature vs. time graph shows how much time it takes to reach constant wire temperature before the cutting starts. The minimum time to reach the stable temperature can be identified. Some amount of foam has to be cut before entering into the required part to get a stable cut

with uniform properties. Since the machine was designed to cut foam for mould making process, the experiment was conducted to analyse the relationship in parameters of Styrofoam only. Further analysis is needed to compare and validate the test results on different materials.

When current increases and feed rate decreases, the kerf width increases. The width of the kerf should be predicted before the cut according to the current and feed rate to compensate the wire path for accomplish the required accuracy and tolerances. It was determined to confine the kerf width to 1.0 mm to get an accurate cutting. The 3D graph of the kerf width with respect to cutting temperature and feed rate was plotted to analyse the relationship between feed rate and temperature to obtain 1mm kerf. It is shown in Fig.10.

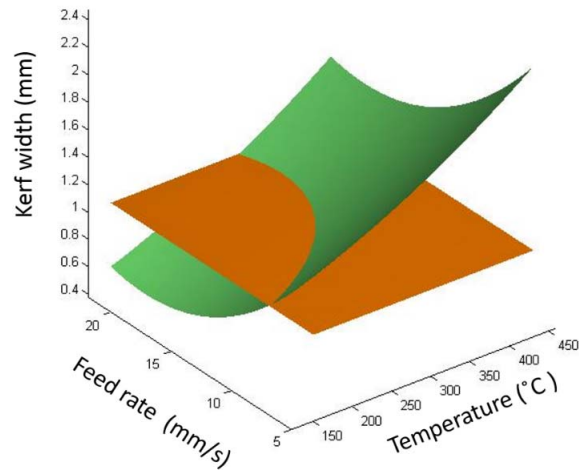


Fig.10 Variation of kerf width with respect to temperature and feed rate

The intersection of the plot with the 1 mm kerf width plane was projected to the XY plane to find the controlling contour of the wire cutting.

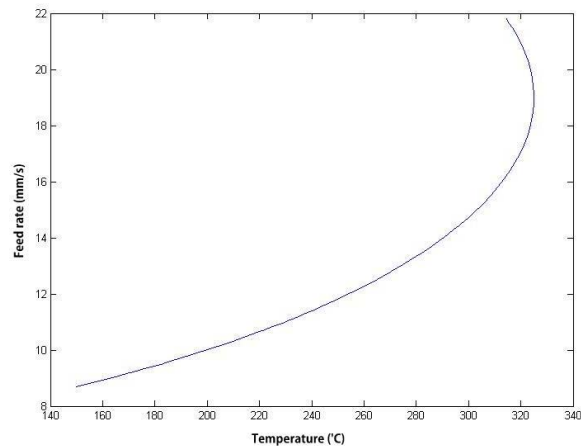


Fig.11 Projected contour on temperature, feed rate plane for 1mm kerf width

From the projected contour shown in Fig.11, the feed rate controlling function to get a 1mm kerf width was identified as $f(x)=0.0005 x^2 -0.1617 x + 23.7694$ (x = temperature $f(x)$ = feed rate) Based on the control equation the required feed rate is linearly proportional to temperatures under 200°C. The previous results enhanced that pure thermal cutting is not applicable on the above temperature range and high level of active controlling of parameters is needed. In contrast the temperature range between 300°C-320°C provides more flexible range of feed rates (15-25 mm/s). The design of the Numerically Controlled machines should be done compatible to above cutting characteristics.

III. CONCLUSION & FUTURE WORK

The hot wire foam cutting is an efficient way of producing foam moulds and different foam shapes. It was clearly observed that a thinner gauge wire gives a small kerf width higher accuracy, while thick wires have high durability. Numerical control methods can be used to control the feed rate and temperature to obtain a constant kerf width with good accuracy. The control equation for the numerically controlled feed rate was derived using the experiment data. The conceptual design of a computer numerically controlled

machine which can perform these tasks is shown in Fig.12.

An active temperature controlling method based on PID control should be incorporated with the design to control the temperature in a limited range to reach high accurate cutting. The drive mechanism of the control axes should be designed by selecting the motors that can provide the appropriate feed rate. The control program should also develop considering the behaviour of characteristics of the particular foam type.

The preliminary study yielded important results about controlling and designing a numerically controlled hot wire cutting with high accuracy and reliability. Further research is recommended choosing the cutting parameter characteristic curves of different foam types.

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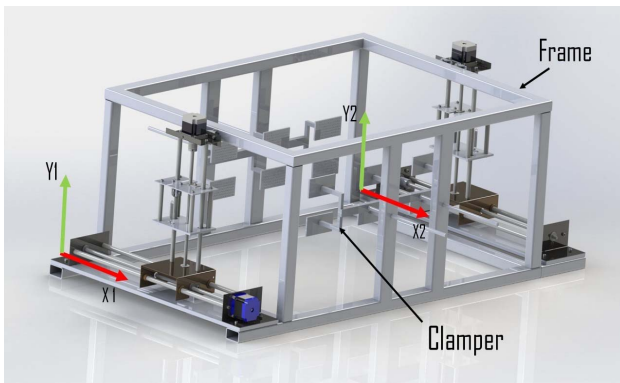


Fig.12 Numerically controlled foam cutting mechanism